Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 2:** Variables on Both Sides

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $4x + 1 = 2x - 5$

$-2x \quad -2x$

$2x + 1 = -5$

$-1 \quad -1$

$\frac{2x}{2} = \frac{-6}{2}$

$\boxed{x = -3}$

2. $y - 3 = -3y - 43$

$+3y \quad +3y$

$4y - 3 = -43$

$+3 \quad +3$

$\frac{4y}{4} = \frac{-40}{4}$

$\boxed{y = -10}$

3. $7 - 14m = 2m - 5$

$+14m \quad +14m$

$7 = 16m - 5$

$+5 \quad +5$

$\frac{12}{16} = \frac{16m}{16}$

$\boxed{\frac{3}{4} = m}$

4. $-10b + 5 = 7b + 5$

$+10b \quad +10b$

$5 = 17b + 5$

$-5 \quad -5$

$\frac{0}{17} = \frac{17b}{17}$

$\boxed{0 = b}$

5. $r + 15 = 4r - 6$

$-r \quad -r$

$15 = 3r - 6$

$+6 \quad +6$

$\frac{21}{3} = \frac{3r}{3}$

$\boxed{7 = r}$

6. $10 - 2v = -5v - 50$

$+5v \quad +5v$

$10 + 3v = -50$

$-10 \quad -10$

$\frac{3v}{3} = \frac{-60}{3}$

$\boxed{v = -20}$

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7. $3(5y + 1) = 18y$

$15y + 3 = 18y$

$-15y \quad -15y$

$3 = 3y$

$\frac{3}{3} \quad \frac{3y}{3}$

$\boxed{1 = y}$

8. $4 - 3x = 8 - 4x$

$+4x \quad +4x$

$4 + x = 8$

$-4 \quad -4$

$\boxed{x = 4}$

9. $8(2w - 2) = 7(3w + 2)$

$16w - 16 = 21w + 14$

$-16w \quad -16w$

$-16 = 5w + 14$

$-14 \quad -14$

$\frac{-30}{5} = \frac{5w}{5}$

$\boxed{-6 = w}$

10. $3(5x + 2) = 2(3x - 6)$

$15x + 6 = 6x - 12$

$-6x \quad -6x$

$9x + 6 = -12$

$-6 \quad -6$

$\frac{9x}{9} = \frac{-18}{9}$

$\boxed{x = -2}$

11. $6(m - 1) = 3(3m + 5)$

$6m - 6 = 9m + 15$

$-6m \quad -6m$

$-6 = 3m + 15$

$-15 \quad -15$

$\frac{-21}{3} = \frac{3m}{3}$

$\boxed{-7 = m}$

12. $7(p + 3) + 9 = 5(p - 2) - 3p$

$7p + 21 + 9 = 5p - 10 - 3p$

$7p + 30 = 2p - 10$

$-2p \quad -2p$

$5p + 30 = -10$

$-30 \quad -30$

$\frac{5p}{5} = \frac{-40}{5}$

$\boxed{p = -8}$

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